

Anonymous Authors¹

ABSTRACT

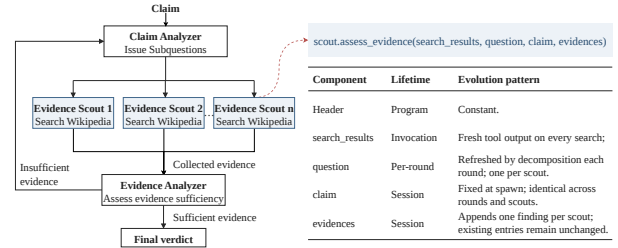
LLM agent frameworks increasingly expose model invocations through structured programming abstractions, with framework runtimes lowering program-level constructs into concrete prompts. We observe that these programmatically generated prompts are inherently heterogeneous: their components exhibit different sharing scopes, lifetimes, and evolution patterns across invocations. However, existing serving systems treat prompts as opaque text, losing this structure and missing opportunities for more effective KV-cache reuse and management. To address the gap, we present ANON, a compiler-serving co-design that recovers and exploits this heterogeneity for efficient KV-cache management.

1 INTRODUCTION

Agent frameworks increasingly expose LLM calls as programming abstractions rather than manual prompt construction. Systems such as ByLLM (Dantamarayana et al., 2025), NVIDIA Object-Oriented Agents (NOOA) (Furgale et al., 2026), and DSPy (Khattab et al., 2024) allow developers to describe model behavior using functions, objects, signatures, and modules, which framework runtimes lower into concrete prompts. This approach also appears in systems such as SIGIL (Dantamarayana et al., 2026), which compiles natural-language skill documents into programs containing typed LLM calls. LLM requests from these frameworks are generated programmatically: they are the serializations of program state rather than free text.

We observed that these programmatically generated LLM requests are inherently heterogeneous. The heterogeneity arises from three dimensions: (1) Sharing scope: one value can be shared across multiple invocations, or only within a single invocation. (2) Lifetime: Different bindings have different lifetimes. Some bindings are single-shot volatile inputs, while others may be reuse across the whole session. (3) Evolution pattern: context for LLM calls can evolve differently overtime, such as remaining fixed, being replaced, or accumulating across invocations. Figure 1 illustrates this heterogeneity in a HoVer-style (Jiang et al., 2020) fact check agent. The claim persists throughout a verification session, the evidence list accumulates over time, and each search result is consumed once and replaced. Although these values appear in the same model invocation, they offer

substantially different reuse opportunities.



```
def assess_evidence(search_results: str, question: str, claim: str, evidences: list[Evidence]) -> Finding by llm()
```

Figure 1. Heterogeneity within a HoVer-style (Jiang et al., 2020) Fact Check Agent’s generated prompt for the summary-making function. It evaluates the sub-question against the accumulated evidence and search results.

This distinction matters because main-stream KV-cache serving backbones are prefix-based. In the example, placing the accumulating evidences binding after the short-lived search_results binding prevents the model from reusing the evidence prefix whenever the search result changes. The model must therefore recompute KV states for tokens whose contents remain unchanged. Moreover, without knowing which components persist or evolve, the serving system cannot accurately determine which cached tokens should be retained, evicted, or prefetched.

However, existing agent serving systems primarily optimize at the level of requests or agent invocation sequences. Pythia (Yu et al., 2026) profiles agent workflows to guide cache management, request scheduling, and autoscaling. CacheScout (Zhang et al., 2026) learns execution transitions online to guide eviction and prefetching, while PBKV (Zheng et al., 2026) predicts future invocations and estimates the reuse potential of cached requests. These sys-

¹ Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

tems exploit predictability between invocations, but treat the contents of each prompt as an opaque token sequence. Consequently, they cannot identify the heterogeneity within a single prompt, leading to a suboptimal serving performance for such program-integrated agents.

To close the gap, we present ANON, a compiler-serving co-design that recovers and exploits prompt heterogeneity for efficient KV-cache management. ANON achieves this by exposing additional prefix reuse and keeping the resulting KV states resident when they are likely to be needed. The identification and exploitation of prompt heterogeneity introduces the following two challenges:

First, heterogeneity is not always explicit in prompt. The sharing scope and lifetime of bindings are closely bonded with the program-level structure and actual runtime assignment. Determining which layout best serves prefix caching is therefore not trivial. ANON combines static analysis with runtime information to determine prompt layouts. The compiler derives a partial order over prompt components based on their sharing scopes and lifetimes. Then, ANON uses the actual token lengths of bindings observed from log to order components left unconstrained by this partial order, producing a concrete layout that improves prefix cache efficiency.

Second, exploiting future reuse requires coordinating cache retention and prefetching. Prompt components differ in when they will be reused and how much recomputation their cached prefixes can avoid. Retaining these prefixes consumes capacity needed by other requests, while prefetching too early occupies device memory and prefetching too late fails to hide transfer latency. Inappropriate KV-cache eviction would lead to unnecessary recomputation. ANON addressed this challenge by combining static analysis hints and runtime observation to instruct retention, eviction and prefetching.

We implement ANON in Jac ByLLM (Mars et al., 2023; Dantanarayana et al., 2025) and use SGLang (Zheng et al., 2024) as the underlying model-serving engine. We evaluate ANON on three representative agent workloads: a HoVer-style (Jiang et al., 2020) fact-checking agent, a HumanEval (Chen et al., 2021) coding agent, and a BFCL (Patil et al., 2025) tool-use agent. We further evaluate a mixed workload comprising all three agent types to assess ANON’s end-to-end performance under concurrent, heterogeneous demand.

In summary, we make the following contributions: (1) We identify the inherent heterogeneity in programmatically generated prompts and its implications for serving. (2) We design and build ANON, a compiler-serving co-design that recovers and exploits prompt heterogeneity for efficient KV-cache management. (3) We conduct comprehensive evaluations of ANON against multiple established serving systems,

demonstrating that ANON achieves up to 48% lower average JCT and 78% lower average TTFT than Vanilla SGLang, and $1.44\times$ throughput on a mixed agent workload.

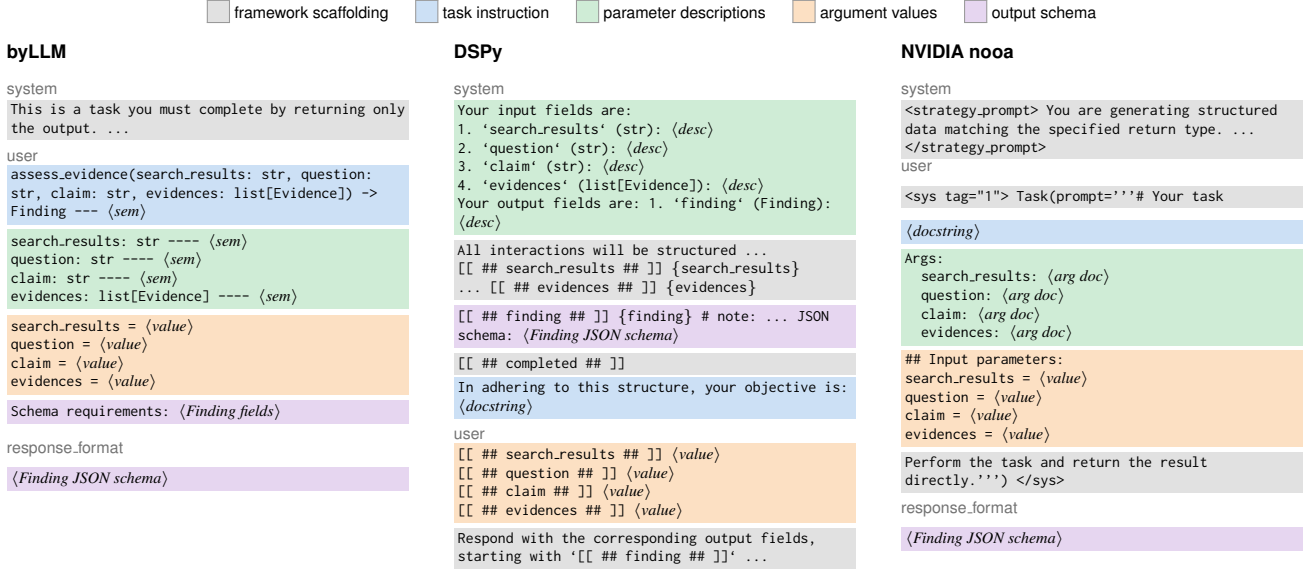
2 BACKGROUND

2.1 Serving systems for LLM agents

LLM serving engines retain the attention keys and values (KV) of tokens they have already processed to avoid repeating computation. PagedAttention manages these states in shareable memory blocks (Kwon et al., 2023), while SGLang indexes cached token sequences in a radix tree (Zheng et al., 2024). An incoming request reuses the longest cached prefix of its prompt and computes KV for the remaining suffix. Reuse stops at the first token mismatch: that token and every subsequent token must be recomputed, even if identical text appears later in a cached sequence, because its preceding context has changed. Prompt Cache supports modular reuse beyond a shared prefix, but requires modules to retain fixed, schema-assigned token positions (Gim et al., 2024).

Retained KV competes with running requests for device memory. Under memory pressure, engines such as SGLang evict unused cached prefixes by recency (Zheng et al., 2024). Hierarchical caching systems, including CachedAttention (Gao et al., 2024), Mooncake (Qin et al., 2025), and SGLang HiCache (SGLang Contributors, 2026), extend the cache beyond device memory with host memory and, optionally, storage. KV evicted from the device can remain in the host tier and be transferred back when needed; prefetching can overlap this transfer with other work. Reloading retained KV can be cheaper than recomputing it. The host tier is also limited: admitting one sequence consumes space that could retain another, so offloading does not eliminate cache-capacity pressure.

An agent *session* is one execution of an agent task, comprising a chain of dependent LLM calls separated by tool and code execution. Between calls, the session’s KV sits idle and is exposed to recency eviction precisely while the agent is doing work that may lead to its reuse. Agent-aware serving systems use execution information to manage these gaps. KVFlow uses an agent step graph to estimate steps to future invocations and guide eviction and prefetching (Pan et al., 2025). Continuum retains KV across tool calls with a time-to-live (Li et al., 2025). CacheScout learns transitions between agents online to guide retention and prefetching (Zhang et al., 2026), while PBKV predicts future agent invocations from workflow history and the current context (Zheng et al., 2026). These systems share a request- or agent-step-level view of future execution: they organize cache management around the reuse of whole requests or agent steps.



`<sem>`, `<docstring>`, `<desc>` and `<arg doc>` are the description strings each framework takes from the program (byLLM sem strings, DSPy Signature docstring and field desc, nooa function docstring and its Args entries); `<value>` is the runtime argument.

Figure 2. The prompt structure of LLM invocation function `assess_evidence`, lowered by three frameworks. They have similar structures: Same five kinds of component, only relative position differs.

2.2 Agent frameworks with LLM calls as programming abstractions

A class of agent frameworks exposes model-backed computation through typed programming interfaces. Examples include byLLM’s meaning-typed programming (MTP) abstraction (Dantanarayana et al., 2025), DSPy’s Signatures and Modules (Khatab et al., 2024), and NOAA’s typed Python methods (Furgale et al., 2026). SIGIL likewise compiles skill descriptions into harnesses containing typed model operations (Dantanarayana et al., 2026). The developer declares a function or signature describing the desired behavior, its inputs, and its return type; invoking it is an ordinary call within the program’s control flow. For example, `assess_evidence` in Figure 1 accepts search results, a question, a claim, and accumulated evidence, and returns a typed finding. We use *call site* to mean a location in the program that invokes such an LLM-backed operation. A session is a concrete execution through these sites; a loop can visit the same site repeatedly with different argument values.

At each invocation, the framework runtime lowers the declaration and its argument values into model-facing messages. The declaration supplies the function name, docstring or semantic strings, parameter types and descriptions, and return schema. This information is known when the program is compiled: byLLM captures it in MTP’s meaning-based intermediate representation, MT-IR (Dantanarayana et al., 2025). The argument values become available only at runtime, when the call executes. Thus, a generated prompt

combines a description fixed by the program with values supplied by that execution.

Some frameworks make deliberate choices about how to arrange this information. NOAA places static context first, appended event history next, and volatile state last to maximize KV-cache reuse across turns (Furgale et al., 2026). DSPy’s adapters use a fixed ordering of system instructions, demonstrations, conversation history, and current inputs; its ChatAdapter emits input fields in signature order (DSPy Contributors, 2026). These layouts follow hand-chosen conventions applied across call sites and programs, rather than an order derived from the properties of a particular program’s bindings. The framework knows the structure behind every prompt, but the serving engine receives only the token sequence.

3 MOTIVATION

Runtime-lowered prompts have similar structures Although these Program-Integrated LLM frameworks differ in surface syntax, they lower LLM invocations in similar ways. They declare LLM invocations as functions or function-like declarations, and framework serialize that declaration together with the runtime arguments into a single prompt.

An example in Figure 2 demonstrate this. ByLLM in Jac, DSPy, NOAA lower the same `assess_evidence` LLM invocation. These prompts carry the same five kinds of components: framework scaffolding, task instruction, parameter descriptions, argument values, and output schema. The only

difference is their relative position. What’s more, argument values are all emitted in declaration order.

The component order carries no intent The order in which these five components appear is decided solely by the framework runtime, without any program specific knowledge. Figure 2 showed the three representative frameworks arrange the same five components differently for the same equivalent invocation, which suggests that no particular order is required by the task. Even when argument values appear in declaration order, they do so simply because the runtime follows the parameter list, whose order was chosen by human developers.

Prompt Components are heterogeneous A prompt programmatically generated by an agent framework is not prose. Instead, it has inherent heterogeneity. The runtime constructs the prompt from different sources, such as program constants, runtime values, external inputs. Therefore these components have different sharing scopes, lifetimes and evolution pattern.

The example in Figure 1 shows this. As we discussed in section 1, this LLM function’s bindings carry such heterogeneity. `claim` is constant, `evidences` accumulates, `question` is one-shot temporary input from other call-sites, `search_results` is external input and only appears once. Blindly emitting these bindings in declaration order therefore blocks prefix cache reuse. Likewise, existing agent serving frameworks treat prompts as uniform token sequences, and therefore cannot effectively retain or release the components based on their future usage.

In summary, the heterogeneity within these programmatically generated prompt exposes new opportunities for efficient serving. However, none of these informations are served for effective serving, which motivates the design of ANON: Recovers heterogeneity by static analysis and use it to unlock more efficient KV-cache scheduling.

4 DESIGN

4.1 Overview

In this section, We present the design of ANON, an efficient serving engine for program-integrated LLM agents. ANON adopts a two-tier compiler-server co-design. One the compiler side, ANON performs static analysis to gather heterogeneity metadata, and on the server side, ANON leverages this metadata to optimize KV-cache scheduling. We first introduce an interface *Agent Descriptor* to bridge the compiler and server, together with the methodology of their construction. Second, we present a heterogeneity-aware KV-cache scheduling strategy that leverages the agent descriptor to enhance serving performance.

4.2 Agent Descriptor for Heterogeneity Metadata

Here we define *Agent Descriptor* to bridge static analysis and server-side scheduling. An agent descriptor contains two parts: (i) **Call-site Template**: It provides a heuristic of the components’ lifetime, sharing scope and evolution pattern. (ii) **Workflow Automaton**: It contains the program-level structure of an agent, including workflow dependencies, execution time estimation and transition probabilities. ANON’s two-tier design let the compiler to provide a skeleton, while the server refines the weight with runtime observations. Two sides are therefore connected by this agent descriptor, and jointly serve a more efficient KV-cache scheduling strategy.

4.2.1 Call-site Template

As discussed in §3, the components of a programmatically generated prompt can be classified into the following two categories: (i) **Invariant Specifications**: the components that are fixed once the program is compiled, such as the system prompt, response schema, and the tool declarations. They are rendered from the compiler’s own intermediate representation and stays constant. (ii) **Dynamic Bindings**: because LLM calls are declared as ordinary functions, the arguments of a call site are supplied in runtime. These runtime values are the dynamic bindings.

We therefore define the call-site template as

$$\mathcal{T}_c = \langle \mathcal{I}_c, \mathcal{B}_c, \mathcal{H}_c \rangle,$$

where $\mathcal{I}_c$ is the invariant specifications, $\mathcal{B}_c$ is the dynamic bindings, and $\mathcal{H}_c$ is the heterogeneity annotation over those bindings. All three are recovered by static analysis.

Heterogeneity annotation. For each dynamic binding, $\mathcal{H}_c$ records both how its value evolves from one call to the next and how long that value stays valid:

$$\mathcal{H}_c : \mathcal{B}_c \longrightarrow \mathcal{E} \times \mathcal{S}.$$

The evolution component $\mathcal{E}$ classifies a binding by its value provenance. We distinguish $\mathcal{E}$ into five classes, listed from the most to the least reusable:

1. **CONST**: the value is fixed by the program text and is identical at every call.
2. **COPY**: the value is the same as the value another call site carries, so the bytes are already resident once that call site has run.
3. **RESP**: the value comes from the reply produced by another call site.
4. **EXTEND**: the binding is an append-only context, and the previous value remains a prefix of the new one.

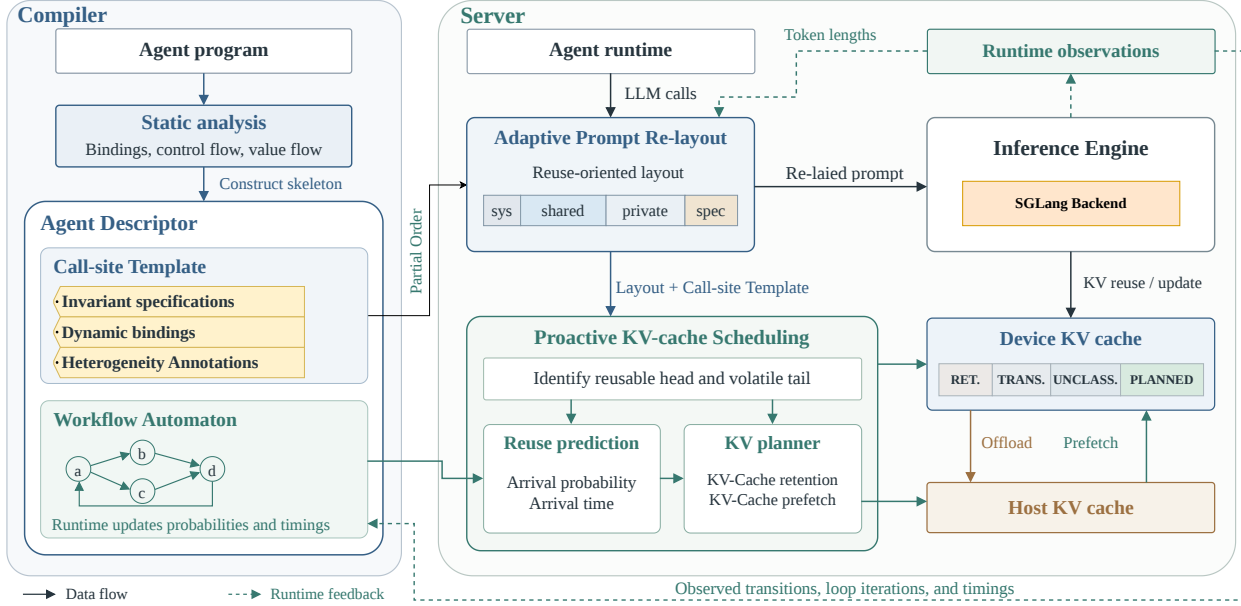


Figure 3. Architure overview of ANON.

5. **VOLATILE**: the value is freshly produced at each call, or its origin lies outside what the analysis can see.

We write ρ for the rank of a class in the order above, which expresses how much of a binding’s bytes an earlier call can already supply.

The lifetime component $\mathcal{S}$ names the program scope whose completion invalidates a binding’s value ($\mathcal{S} = \emptyset$ meaning that the value survives for the whole session once produced). For example, `claim` in `scout.assess_evidence` in figure 1 has $\mathcal{E} = \text{CONST}$ and $\mathcal{S} = \emptyset$. Lifetime is independent of evolution, and the server needs both. The evolution class governs which bytes a later prompt can reuse, while the lifetime governs when cached bytes can be given up.

4.2.2 Workflow Automaton

An agent typically consists of a sequence of dependent LLM calls whose workflow structure is implicit in the program. Concretely, we abstract an agent as a time-annotated probabilistic finite-state machine (TPFM). We formulate the TPFM as:

$$\mathcal{A} = (Q, P, q_0, F, \mathcal{T}),$$

where:

1. Q is the set of states. Each state $q \in Q$ is labelled by an LLM callsite;
2. P denotes the transition probabilities between states.
3. $q_0 \in Q$ is the entrance callsite.

4. F specifies the termination probability of the agent at each state.

5. $\mathcal{T} = \{D_q^{\text{exec}}, D_{q,q'}^{\text{gap}}\}$ denotes the timing annotations associated with states and transitions, where D_q^{exec} is the execution-time distribution of the LLM callsite represented by state q , and $D_{q,q'}^{\text{gap}}$ is the inter-call gap distribution between the completion of callsite q and the invocation of its successor q' .

Static analysis fixes the shape of this automaton, while its weights comes from runtime logs. The states, the entry, the transitions among them, and final states are recovered from code. $\mathcal{T} = \{D_q^{\text{exec}}, D_{q,q'}^{\text{gap}}\}$ and P are often unknown at compile time and model dependent. The server observes these online and accumulates them across sessions, updating the statistical model of the agent consistently. Together the two halves give the server a forward-looking view of a session.

4.3 Heterogeneity-aware KV-cache Arrangement

4.3.1 Adaptive Prompt re-layout

Because prefix cache reuse stops at the first byte that differs. The message a framework emits for a call site follows the declaration order of the function, therefore may disable the potential of prefix reuse. ANON’s adaptive re-layout permutes the components of each call message so that the bytes most likely to be resident already come first, which enables an improved prefix reuse. The layout permutation’s

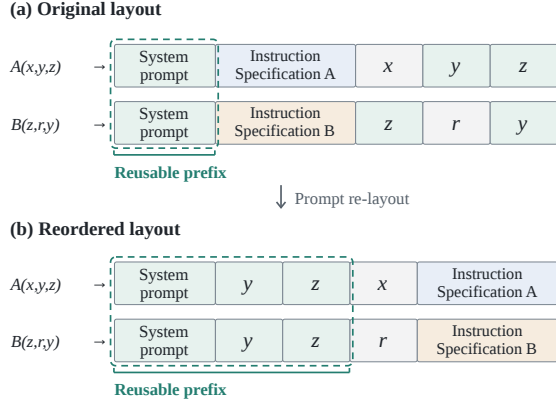


Figure 4. Example of adaptive re-layout, which increases prefix reuse by moving the more reusable components to the front, exposing shared prefix to the KV-cache.

basic is the compiler derives a partial order over components, as digested in template $\mathcal{T}_c$ 4.2.1, server then freezes the re-layout verdict with observed token length from runtime.

Transportable Components First, the dynamic bindings $\mathcal{B}_c$ can be permuted freely. The compiler lowers each argument into a self-contained component labelled by its parameter name, and the framework places these components in declaration order only because that is the order of the argument list; Re-layout may therefore choose any order of the bindings.

Second, part of the invariant specification $\mathcal{I}_c$ is movable as well. $\mathcal{I}_c$ is a simplified abstraction. More precisely, it consists system prompt, reponse schema, tool schema and instruction specification(Task description and parameter description generated by compiler). Among them, system prompt stays universally constant, and two schemas are digested by model separately. Instruction header is movable.

ANON therefore operates on the order of bindings and the placement of instruction specification. The following paragraphs describe how ANON chooses a layout within this space.

Layout decision algorithm The layout of a call site is a program-wide decision made from the compiler-derived template $\mathcal{T}_c$. The template alone yields only a partial order, so the server observes the token length of each binding’s value at runtime to break the ties within each band. Concretely, the bindings of a call site are sorted in ascending lexicographic order of the tuple

$$K(b) = (\text{shared}, \text{scoped}, \rho, \text{depth}, -\text{length}),$$

whose components are defined as follows:

- *shared*: whether the value of b is carried by more than one call site; 0 if shared and 1 otherwise, so shared bindings lead.
- *scoped*: the lifetime of b ; 0 if it is session-wide ($\mathcal{S} = \emptyset$) and 1 otherwise, so bindings that survive scope boundaries precede those reset by them.
- ρ : the stability rank defined in §4.2.1, from 1 for CONST to 5 for VOLATILE, so the more reusable class leads.
- *depth*: the number of transitions in $\mathcal{A}$ (§4.2.2) from q_0 to the call site at which the value of b first appears, so values that enter a session earlier lead and the prompt of an upstream call site becomes a prefix of those downstream.
- *length*: the token length of the value of b , observed by the server; negated so that longer bindings stay in front and a change in a shorter binding invalidates fewer cached tokens.

The first four components are fixed by the program and constitute the compiler’s partial order; only *length* is observed in at runtime. For a binding shared by several call-sites, the value of *scoped* and ρ will be evaluated among all call-sites that carry it to make sure consistency of ordering across call-site. Specifically, the value is marked as *scoped* if any of the call-sites bounds its lifetime, and its stability ranking is **EXTEND** if any of them appends to it and **COPY** otherwise.

specification placement. The *instruction specification* is lowered from code by compiler, which is unique for each call-site. Consequently, emitting it first would prevents prefix sharing among different call-sites, however many values they have in common. After sorting, ANON moves the specification behind the bindings whenever the leading binding is shared, so that sibling call sites can share large blocks of common bindings. A per-program switch is additionally designed to keeps the specification in front when moving it is suspected of changing the model’s behavior.

As shown in Algorithm 1, ANON first computes the sorting tuple of every binding, then performs a per call-site sort to obtain its final layout.

4.3.2 Proactive KV-cache Scheduling

While adaptive prompt re-layout improves prefix reuse only by reshaping the request, ANON’s KV-cache scheduling proactively proactively arrange the placement of KV cache entries. Using the agent descriptor, the server decides for every served KV-cache sequence what to keep, what to give up, and what to load back ahead of time.

Algorithm 1 Layout decision Program

Require: Per call-site templates $\{\mathcal{T}_c\}$;
observed lengths ℓ ;
specification-pinning switch pin ;
Ensure: Layout L_c for each call site c

- 1: **for all** $b \in \bigcup_c \mathcal{B}_c$ **do**
- 2: $K(b) \leftarrow (shared, scoped, \rho, depth, -\ell(b))$
 $\triangleright$ Assign sorting tuple for each binding
- 3: **end for**
- 4: **for all** call sites c **do**
- 5: $\pi_c \leftarrow \text{SORT}(\mathcal{B}_c, K)$ $\triangleright$ ascending lexicographic
- 6: $s_c \leftarrow \text{instruction specification from } \mathcal{I}_c \in \mathcal{T}_c$
- 7: **if** $\neg pin \wedge (\exists b \in \mathcal{B}_c : K(b)[shared] = 0)$ **then**
- 8: $L_c \leftarrow \pi_c || \langle s_c \rangle$ $\triangleright$ Expose the shared prefix
- 9: **else**
- 10: $L_c \leftarrow \langle s_c \rangle || \pi_c$ $\triangleright$ Keep the specification first
- 11: **end if**
- 12: **end for**
- 13: **return** $\{L_c\}_c$

Identify Reusable KV-cache Since re-layout already places the stable and shared bindings at the front, a served prompt naturally divides into two segments: a reusable head and a volatile tail. In a full prompt, what behind the reusable head are all classified into volatile tail.

Reusable head consists two parts: a *static prefix* and a *session-shared extension*:

- **Static Prefix** is what the call site emits without run-time inputs, and its values outlives single session. It includes the invariant specifications $\mathcal{I}_c$ together with the CONST bindings that re-layout has placed right behind them.
- **Session-shared Extension** is the run of bindings behind the static prefix that a later call of the same session will emit again at the same position.

The two parts are derived from agent program itself. $\mathcal{H}_c$ names the provenance of each binding, and the automaton $\mathcal{A}$ names the transitions between call sites. Additionally with the decided layout of each call-site, we determine the reusable head by jointly analyze them.

Predict Future Reuse Because the workflow automaton $\mathcal{A}$ already exposes all potential next calls, our prediction focus on the following three aspects: (i) The probability of the arrival. (ii) When will the call arrive. (iii) Which prefix of its prompt is already known.

Probability of arrival. We estimate the probability of the next call by a loop-tagged table. As the compiler side emits the workflow graph, ANON adopts Bourdoncle-style (Bour-

doncle, 1993) weak topological ordering to discover all loops within the agent, so that each call-site q is assigned the chain of loops enclosing it. The server counts the iterations of each loop per session, so every call carries a tuple ζ of the current iteration numbers of the loops enclosing its call-site, outermost first. We refine the transition and termination probabilities of $\mathcal{A}$ by this tuple and define the table as

$$\mathcal{P}(q' | q, \zeta) \quad \mathcal{F}(q, \zeta)$$

where ζ is the loop indices; $\mathcal{P}$ stands for the probability that a call-site q in context ζ is followed by a call-site q' ; $\mathcal{F}$ stands for the probability that the session ends at call-site q under context ζ . With an empty ζ they reduce to P and F of §4.2.2.

The table is a frequency count that the server updates across sessions. When a call arrives, its call-site is counted as the outcome of the previous call-site under the previous call's ζ , and termination is counted when the session closes. Each observation is also counted under every suffix of ζ down to the empty tuple, so that for the same q the table holds one row per level of context, from the fully specified iteration to the pooled count over all iterations. After a call at q lands, we compute the probability of each outcome o by Witten–Bell interpolation from the empty context ζ_0 to the full context $\zeta_m = \zeta$, where o is a successor of q or a session termination:

$$\hat{P}_k(o) = \lambda_k \frac{c_k(o)}{n_k} + (1 - \lambda_k) \hat{P}_{k-1}(o),$$
$$\hat{P}_0(o) = \frac{1}{|\text{succ}(q)|}, \quad \lambda_k = \frac{n_k}{n_k + d_k},$$

where $\text{succ}(q)$ is the set of static successors of q (including termination, which is treated as at an exit call-site). ζ_k is the suffix of ζ that keeps only the innermost k indices, and the three counts are read from the row (q, ζ_k) of the table: $c_k(o)$ is how many times outcome o followed a call at q in context ζ_k , $n_k = \sum_o c_k(o)$ is the number of observations in that row, and d_k is the number of distinct outcomes with $c_k(o) > 0$.

Then,

$$\mathcal{P}(q' | q, \zeta) = \hat{P}_m(q') \quad \mathcal{F}(q, \zeta) = \hat{P}_m(\text{end})$$

A level with no observation is skipped. The weight λ_k grows with the support of a level, so a well-observed iteration dominates the prediction while a sparse one defers to the pooled behavior of q .

Estimate call arrival time. The time of arrival is conservatively approximated based on record statistics. The workflow automaton records $D_{q,q'}^{\text{gap}}$ per transition: each time a call at q' arrives, the server appends the interval between the reply of the preceding call at q and this arrival. The

interval is the agent’s own time spent in its code between two calls.

We estimate the gap before the next call as:

$$\hat{g} = t_{50} - \frac{1}{2}(t_{90} - t_{50})$$

where t_{50} and t_{90} are the 50th and 90th percentiles of the gap samples on the transition (q, q') ; the predicted call is then expected $\hat{g}$ after the reply of the current call. We introduce the second term to make sure a transition with more sparse distribution of gap receives a tighter estimation.

Construct prompt ahead of time. As indicated in 4.3.2, the prompt of the known successor q' is reconstructible, or at least partial-reconstructible. ANON builds it prompt in the layout $L_{q'}$, using the values that the session already carries. The construction processes bindings in the order of $L_{q'}$ while consulting their evolution class, deducing value accordingly. The construction stops at the first binding with no known origin. Then the matched prefix is sent to kv-planner for speculative preparation.

Lifecycle aware KV-cache scheduling As we discussed in 3, different components have different lifetime, sharing scope and evolution patterns. Therefore, relying on recency based LRU would lead to suboptimal performance. ANON adopts a banded priority and steers KV-cache upon it.

Definition of Priority Bands. We categorize KV-cache entries into four bands, from first evicted to last:

1. **RETIRED:** the volatile tail of prompt belonging to completed sessions.
2. **TRANSIENT:** the volatile tail of a served prompt of a live session. The session that produced them will not present them again.
3. **UNCLASSIFIED:** entries that analysis cannot categorize, such as the prompts of a program that was never registered. They keep the engine’s default treatment.
4. **PLANNED:** prefixes that a predicted call will reuse. Inside this band entries are ordered by the expected value of keeping them resident, defined next.

Eviction empties the lowest band first and falls back to recency only within a band. Generally, KV-cache that will likely be reused left memory later.

Band Based KV-cache scheduling. As we mentioned above, eviction empties the lowest band first and falls back to recency only within a band, so KV-cache that will likely be reused left memory later. On the host tier, however, it keeps its recency order except that retired segments go first, so a head the device gave up under pressure remains

reloadable while it may still be needed. Additionally, KV-cache entries marked as *TRANSIENT* won’t be admitted into host as they rival memory space with reusable KV-cache entries.

Bands are assigned at three moments of a session’s lifecycle. (1) **When a served prompt enters the cache**, ANON splits it at its reusable head (Section 4.3.2). The volatile tail and the reply generated after it become *TRANSIENT*; the head stays *UNCLASSIFIED* until the next prediction refresh finds a successor call that starts with it, which promotes it to *PLANNED*. (2) **When a later call arrives**, the transition it takes may rewrite bindings that earlier heads carried. Each affected head is cut back to the prefix that survives the rewrite, and the bytes past it are demoted to *TRANSIENT*. (3) **When the session ends**, the volatile tails of every prompt it served drop to *RETIRED*. Their heads stay *UNCLASSIFIED* and age out by recency, since other sessions of the same program may still share them.

These bands are mirrored directly into the radix tree’s eviction order. The band is the primary key of the eviction heap and recency breaks ties.

Prediction-based KV-cache protection Based on prediction, kv-planner of ANON conducts protection. The duty of it is to ensure required prefix resides on the device, or at least on host. Each prediction is summarized into a tuple of reconstructed prefix, arrival probability and estimated deadline:

$$Prediction = (\pi_{q'}, \mathcal{P}(q' | q, \zeta), t_{reply} + \hat{g})$$

where $\pi_{q'}$ is the prefix reconstructed for q' and t_{reply} is the time the current call replied. Predictions are refreshed at every call, and are dropped when the predicted calls arrive or pass their deadline. Protection acts on this tuple in two ways. *Retention* keeps $\pi_{q'}$ from leaving the device, and *prefetch* restores the part of $\pi_{q'}$ that has already been demoted to the host.

Retention. The planner ranks the predicted reusable KV-cache entries by the value of keeping $\pi_{q'}$ resident at every refresh:

$$\mathcal{D}(q') = \frac{\mathcal{P}(q' | q, \zeta) \cdot |\pi_{q'}|_{cached}}{\max(t_{reply} + \hat{g} - t_{now}, \tau_{min})}$$

where $|\pi_{q'}|_{cached}$ is the number of tokens of $\pi_{q'}$ cached(both on host and device) and τ_{min} is a floor that keeps a call due now from dominating. ANON’s kv-planner sort all predictions by $\mathcal{D}$ in descending order, and admits them into *PLANNED* until the band covers a fixed fraction of the device tier. Within the band, their $\mathcal{D}$ add up on every radix cache node, making shared head outranks a private one.

Prefetch. When part of $\pi_{q'}$ has been demoted to the host, the planner prefetches it back to the device ahead of the estimated arrive time. A prefetch release deadline is therefore

defined as:

$$t_{\text{release}} = t_{\text{reply}} + \hat{g} - \frac{|\pi_{q'}|_{\text{host}}}{B_{\text{h2d}}} - \frac{|\pi_{q'}|_{\text{uncached}}}{B_{\text{prefill}}}$$

where B_{h2d} and B_{prefill} are the measured host-to-device copy rate and prefill rate. Among released predictions the earliest deadline goes first, and break ties by expected saving:

$$\text{Saving} = \mathcal{P}(q' | q, \zeta) \cdot |\pi_{q'}|_{\text{cached}}$$

A prefetch takes only free device space after reclaiming *RETIRED* and *TRANSIENT* entries, so it never interferes with real requests; when no space is free, prefetching pauses briefly but still prevent predicted KV-cache entries from being completely evicted. A prefetch is released when the engine is idle or its deadline t_{release} approaches.

5 IMPLEMENTATION

We implemented ANON on top of Jac byLLM and SGLang. The serving layer of ANON sits upon model engine and ensure KV-Cache entries are scheduled with respect to their heterogeneity. The compiler side static analysis construct essential metadata from compiler-IR, sending the them to serving layer of ANON. Necessary modifications of SGLang HiCache engine for ANON’s KV-cache arrangement are also made based on demand.

6 EVALUATION

In this section, we evaluate ANON on multiple representative agent workloads, and ablate each components.

6.1 Environmental Settings

6.1.1 Testbed and Model

We conduct all experiments on a single NVIDIA RTX 3090 GPU using Qwen3-14B-AWQ. We allocate 5.2 GB of GPU memory to the device KV-cache pool and 10 GB of CPU memory to the host KV-cache pool.

6.1.2 Baselines

We compare ANON against the following serving systems and ablated variants. (i) **Vanilla SGLang** (Zheng et al., 2024) uses SGLang’s HiCache with LRU eviction. (ii) **CacheScout** (Zhang et al., 2026) identifies agents from their request prefixes and learns inter-agent execution transitions online. It uses the learned transition patterns to prioritize cached KV entries and prefetch the prefixes of predicted successor agents. (iii) **Continuum** (Li et al., 2025) retains each agent program’s KV cache for an adaptively selected time-to-live (TTL). The TTL is derived from observed inter-invocation delays and expected cache reuse, allowing the system to preserve reusable context while the

agent executes tools. (iv) **ANON-KV-only** enables proactive KV-cache scheduling but disables adaptive prompt re-layout. (v) **ANON-Relayout-only** enables adaptive prompt re-layout but replaces proactive KV-cache scheduling with LRU eviction.

6.1.3 Workloads

We evaluate ANON using three representative agentic workloads ported to ByLLM (Dantamarayana et al., 2025) in Jac: HoVer-style (Jiang et al., 2020) fact check agent, HumanEval (Chen et al., 2021) code agent, and a BFCL (Patil et al., 2025) tool-use agent.

Additionally, we conducted a mixture of these workloads to comprehensively verify the performance of ANON. The mixed workload is constructed with 4 lanes of fact check agent, 4 lanes of BFCL tool-use agent and 12 lanes coding agent. Whenever a lane completes its task, it picks another one and launch the new task immediately. The ratio of 1:1:3 is selected deliberately for balancing cache pressure so that three agent workloads have similar KV footprint.

6.2 End-to-End Performance of ANON

We tested ANON under different number concurrent workflows, and a mixed workloads. Across these workloads, we compare average job completion time(JCT), average time-to-first-token(TTFT) and throughput(session/min) for the mixed workload.

Performance on mixed workload As shown in table 2, ANON raises throughput to $1.44\times$ that of Vanilla SGLang and reduces average TTFT and JCT by 34% each. Continuum and CacheScout perform close to Vanilla SGLang on all three metrics. The gain of ANON comes from a higher cache hit rate in both device and host memory.

Average JCT Across the three workloads, ANON reduces average JCT by 6–48% compared with Vanilla SGLang (Figure 5), and by 34% on the mixed workload (Table 2). When concurrency is low, adaptive prompt re-layout accounts for nearly all of the gain. As concurrency grows, re-layout alone stops paying off, and proactive KV-cache scheduling carries the gain: it evicts volatile tails first and protects the reusable heads that predicted calls will need. On fact check, scheduling contributes from the lowest concurrency.

Except for an 8–13% reduction on BFCL at $c = 4$, Continuum and CacheScout perform close to Vanilla SGLang. Both exploit predictability between invocations but treat each prompt as an opaque token sequence. In the original prompt order, volatile bindings often precede reusable ones, making prefixes hold little that later calls can reuse. ANON-KV-only, which schedules on the original prompt order, shows the same limit.

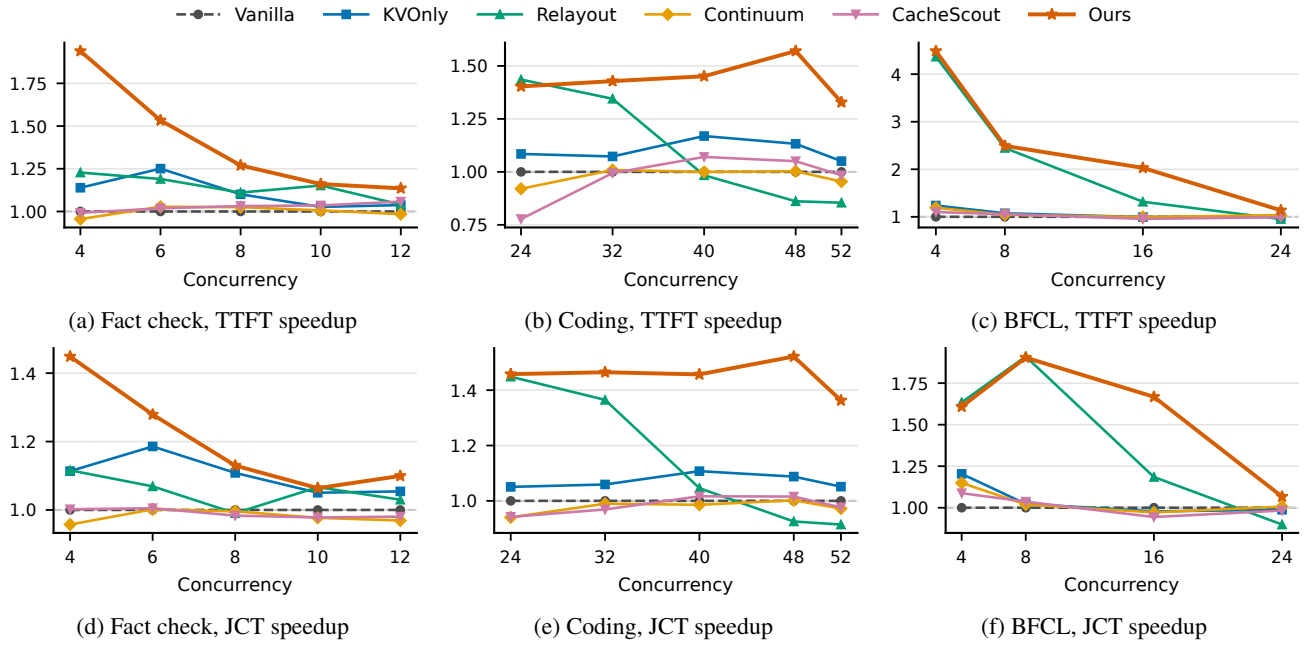


Figure 5. Average TTFT speedup (top) and average JCT speedup (bottom) over Vanilla SGLang under varying numbers of concurrent workflows.

Average TTFT Across the three workloads, ANON reduces average TTFT by 12–78% compared with Vanilla SGLang (Figure 5), and by 34% on the mixed workload (Table 2). As with JCT, adaptive prompt re-layout accounts for nearly all of the gain at low concurrency, and proactive KV-cache scheduling carries it as concurrency grows. On fact check and BFCL, the TTFT reduction exceeds the JCT reduction, because a reused prefix skips its prefill, whereas JCT also includes decoding and tool execution which the KV-cache does not shorten. Continuum and CacheScout show the same limit as in JCT.

Table 1. p -values of a paired test against Vanilla SGLang.

System	Fact check	Coding	BFCL
CacheScout	1.00	0.11	1.00
Continuum	1.00	0.23	1.00
ANON-KV-only	1.00	0.95	1.00
ANON-Relayout-only	1.00	0.19	1.00
ANON	1.00	0.51	1.00

Task Quality Table 1 tests the task quality of each arm against Vanilla SGLang. $p > 0.05$ indicates no significant difference in task quality. None of the arms differs significantly. Performance gains of ANON don’t come at the cost of task quality.

Table 2. Performance on the mixed workload. TTFT and JCT are weighted average over three agents by their lane count. Throughput is calculated under steady load.

System	Throughput (sess/min)	TTFT (s)	JCT (s)	Cache statistics (%)		
				Device	Host	Miss
Vanilla SGLang	2.81	17.5	555	11.3	1.3	87.4
CacheScout	2.81	17.1	543	13.1	0.3	86.6
Continuum	2.81	17.3	552	11.6	1.4	87.0
ANON	4.04	11.5	364	17.3	33.4	49.3
w/o prefetch	3.71	12.3	382	16.7	32.6	50.6
Relayout-only	3.31	14.1	435	8.7	29.5	61.9
KV-only	3.07	16.3	514	14.1	7.2	78.7

6.3 Cache Efficiency of ANON

Figure 6 shows the cache hit rate of prompt tokens at the same concurrency levels as Figure 5. On coding and BFCL, adaptive prompt re-layout alone brings the hit rate at low concurrency to that of ANON: it places the stable and shared bindings at the front, where they form a reusable head that later calls hit. As concurrency grows, the hit rate of ANON-Relayout-only drops quickly, because LRU eviction does not distinguish reusable heads from volatile tails. With proactive KV-cache scheduling, ANON keeps a higher hit rate than ANON-Relayout-only as concurrency grows, and on coding its hit rate stays nearly flat. On fact check, ANON’s hit rate exceeds that of ANON-Relayout-only from the low-

	Fact check					Coding					BFCL				
Vanilla	30	15	11	8	8	23	20	20	20	20	13	9	9	8	
Continuum	27	15	12	8	8	22	21	20	21	22	14	9	9	8	
CacheScout	31	16	12	10	9	22	22	21	21	21	17	10	9	8	
KVOnly	36	29	20	14	12	27	27	27	25	25	14	9	8	8	
Relayout	46	32	28	23	20	57	54	28	15	14	69	64	36	12	
Ours	62	46	35	27	25	56	57	57	56	51	69	64	58	25	
	4	6	8	10	12	24	32	40	48	52	4	8	16	24	
	Concurrency														

Figure 6. Cache hit rate (%) of prompt tokens under varying numbers of concurrent workflows.

est concurrency. Continuum and CacheScout stay close to Vanilla SGLang at every concurrency. ANON-KV-only stays well below ANON, since under the original prompt order the reusable head it can protect is short. The mixed workload shows the same pattern (Table 2).

6.4 Ablation Studies

We ablate ANON on the mixed workload by enabling its components one at a time (Table 2). Besides ANON-KV-only and ANON-Relayout-only, ANON w/o prefetch keeps adaptive prompt re-layout and proactive KV-cache scheduling but disables prefetch. Figures 5 and 6 show ANON-KV-only and ANON-Relayout-only across concurrency levels.

Adaptive prompt re-layout alone reduces JCT by 22% compared with Vanilla SGLang and raises throughput to $1.18\times$. Adding proactive KV-cache scheduling without prefetch reduces JCT by a further 12% and raises throughput to $1.32\times$. Prefetch reduces JCT by another 5% and brings throughput to $1.44\times$.

The two mechanisms depend on each other. Applied to the original prompt order, proactive KV-cache scheduling reduces JCT by only 7% (ANON-KV-only); applied after re-layout, it reduces JCT by 16% relative to ANON-Relayout-only. The combined 34% reduction is larger than the sum of the two individual reductions. Re-layout places the stable and shared bindings at the front, so a served prompt divides into a reusable head and a volatile tail, and scheduling gets a reusable head to protect.

7 RELATED WORK

KV-cache reuse PagedAttention (Kwon et al., 2023) and SGLang (Zheng et al., 2024) reuse KV along an exact shared prefix, and hierarchical caches (Gao et al., 2024; Qin et al., 2025; SGLang Contributors, 2026) extend its capacity beyond device memory. Prompt Cache (Gim et al., 2024) and CacheBlend (Yao et al., 2025) reuse KV beyond the prefix, by fixing module positions in a schema or by selectively recomputing tokens of concatenated chunks. ANON keeps

exact prefix matching and instead changes the prompt so that reusable components form the prefix; it is orthogonal to position-independent reuse.

Existing agent serving systems KVFlow (Pan et al., 2025), Continuum (Li et al., 2025), CacheScout (Zhang et al., 2026), PBKV (Zheng et al., 2026), and Pythia (Yu et al., 2026) predict future execution from step graphs, observed tool-call delays, learned transitions, or workflow histories, and use these predictions to retain and prefetch KV between calls.

Existing Exploitation of prompt structure Agent frameworks (Dantamarayana et al., 2025; Khattab et al., 2024; Furgale et al., 2026; Dantamarayana et al., 2026) know the structure of the prompts they generate. Where they arrange it for reuse, they apply one hand-chosen convention to every call site, such as NOAA’s static, history, and volatile regions. Parrot (Lin et al., 2024) passes application structure to the server through Semantic Variables that developers declare, but serves prompts in the layout the application wrote. SGLang’s frontend (Zheng et al., 2024) reorders prompt statements to lengthen the constant prefix, but has GPT-4 choose the order, which its authors note is not yet reliable.

8 CONCLUSION

The inherent structure and heterogeneity of programmatically generated prompts create opportunities for optimization. ANON identifies and exploits these opportunities rather than treating each prompt as a monolithic unit, thereby unlocking proactive optimizations that improve performance and efficiency. Our evaluation demonstrates that ANON’s methodology successfully creates a more efficient LLM serving system.

REFERENCES

- Bourdoncle, F. Efficient chaotic iteration strategies with widenings. In *Formal Methods in Programming and Their Applications*, volume 735 of *Lecture Notes in Computer Science*, pp. 128–141. Springer, 1993. doi: 10.1007/BFb0039704.
- Chen, M., Tworek, J., Jun, H., Yuan, Q., Pinto, H. P. d. O., Kaplan, J., Edwards, H., Burda, Y., Joseph, N., Brockman, G., Ray, A., Puri, R., Krueger, G., Petrov, M., Khlaaf, H., Sastry, G., Mishkin, P., Chan, B., Gray, S., Ryder, N., Pavlov, M., Power, A., Kaiser, L., Bavarian, M., Winter, C., Tillet, P., Such, F. P., Cummings, D., Plappert, M., Chantzis, F., Barnes, E., Herbert-Voss, A., Guss, W. H., Nichol, A., Paino, A., Tezak, N., Tang, J., Babuschkin, I., Balaji, S., Jain, S., Saunders, W., Hesse, C., Carr, A. N., Leike, J., Achiam, J., Misra, V., Morikawa, E., Radford,

- A., Knight, M., Brundage, M., Murati, M., Mayer, K., Welinder, P., McGrew, B., Amodei, D., McCandlish, S., Sutskever, I., and Zaremba, W. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.
- Dantanarayana, J. L., Kang, Y., Sivasothynathan, K., Clarke, C., Li, B., Kashmira, S., Flautner, K., Tang, L., and Mars, J. Mtp: A meaning-typed language abstraction for ai-integrated programming. *Proceedings of the ACM on Programming Languages*, 9(OOPSLA2):1176–1204, 2025. doi: 10.1145/3763092.
- Dantanarayana, J. L., Kashmira, S., Tang, L., and Mars, J. SIGIL: Compiling agent skills into typed harnesses, 2026. URL <https://arxiv.org/abs/2607.27309>.
- DSPy Contributors. ChatAdapter: DSPy api reference. Online documentation, 2026. URL <https://dspy.ai/ap i/adapters/ChatAdapter/>. Accessed September 14, 2026.
- Furgale, P., Klingler, S., Nolan, J., Staats, M., Di Lorenzo, G., Martinez Abad, E., Schüller, C., Dinu, R., Devoto, A., Berard, P., Kaplun, G., Sarafian, E., Roveri, R., Derzynski, L., and Silveira Cabral, R. NVIDIA-labs OO Agents: Native Python Object-Oriented Agents, 2026. URL <https://arxiv.org/abs/2607.20709>.
- Gao, B., He, Z., Sharma, P., Kang, Q., Jevdjic, D., Deng, J., Yang, X., Yu, Z., and Zuo, P. Cost-efficient large language model serving for multi-turn conversations with CachedAttention. In *2024 USENIX Annual Technical Conference (USENIX ATC 24)*, pp. 111–126, Santa Clara, CA, 2024. USENIX Association. URL <https://www.usenix.org/conference/atc24/presentation/gao-b in-cost>.
- Gim, I., Chen, G., seob Lee, S., Sarda, N., Khandelwal, A., and Zhong, L. Prompt Cache: Modular attention reuse for low-latency inference. In *Proceedings of Machine Learning and Systems*, volume 6, pp. 325–338, 2024. URL https://proceedings.mlsys.org/paper_files/paper/2024/hash/a66caa1703fe34705a4368c3014c1966-Abstract-Conference.html.
- Jiang, Y., Bordia, S., Zhong, Z., Dognin, C., Singh, M., and Bansal, M. HoVer: A dataset for many-hop fact extraction and claim verification. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pp. 3441–3460, 2020.
- Khattab, O., Singhvi, A., Maheshwari, P., Zhang, Z., Santhanam, K., Vardhamanan, S., Haq, S., Sharma, A., Joshi, T. T., Moazam, H., Miller, H., Zaharia, M., and Potts, C. DSPy: Compiling Declarative Language Model Calls into State-of-the-Art Pipelines. In *International Conference on Learning Representations*, 2024.
- Kwon, W., Li, Z., Zhuang, S., Sheng, Y., Zheng, L., Yu, C. H., Gonzalez, J. E., Zhang, H., and Stoica, I. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles*, pp. 611–626, New York, NY, USA, 2023. Association for Computing Machinery. doi: 10.1145/3600006.3613165. URL <https://arxiv.org/abs/2309.06180>.
- Li, H., He, R., Mang, Q., Zhang, Q., Mao, H., Chen, X., Zhou, H., Zhang, H., Cheung, A., Gonzalez, J., and Stoica, I. Continuum: Efficient and robust multi-turn llm agent scheduling with kv cache time-to-live, 2025.
- Lin, C., Han, Z., Zhang, C., Yang, Y., Yang, F., Chen, C., and Qiu, L. Parrot: Efficient serving of LLM-based applications with semantic variable. In *18th USENIX Symposium on Operating Systems Design and Implementation (OSDI 24)*, pp. 929–945. USENIX Association, 2024.
- Mars, J., Kang, Y., Daynauth, R., Li, B., Mahendra, A., Flautner, K., and Tang, L. The jaseci programming paradigm and runtime stack: Building scale-out production applications easy and fast. *arXiv preprint arXiv:2305.09864*, 2023.
- Pan, Z., Patel, A., Hu, Z., Shen, Y., Guan, Y., Li, W.-L., Qin, L., Wang, Y., and Ding, Y. KVFlow: Efficient prefix caching for accelerating LLM-based multi-agent workflows, 2025. URL <https://arxiv.org/abs/2507.07400>.
- Patil, S. G., Mao, H., Yan, F., Ji, C. C.-J., Suresh, V., Stoica, I., and Gonzalez, J. E. The berkeley function calling leaderboard (BFCL): From tool use to agentic evaluation of large language models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 48371–48392. PMLR, 2025.
- Qin, R., Li, Z., He, W., Cui, J., Ren, F., Zhang, M., Wu, Y., Zheng, W., and Xu, X. Mooncake: Trading more storage for less computation—a KVCache-centric architecture for serving LLM chatbot. In *23rd USENIX Conference on File and Storage Technologies (FAST 25)*, pp. 155–170, Santa Clara, CA, 2025. USENIX Association. URL <https://www.usenix.org/conference/fast25/presentation/qin>.
- SGLang Contributors. HiCache system design and optimization. SGLang documentation, 2026. URL https://docs.sglang.io/docs/advanced_features/hicache_design. Accessed September 22, 2026.
- Yao, J., Li, H., Liu, Y., Ray, S., Cheng, Y., Zhang, Q., Du, K., Lu, S., and Jiang, J. CacheBlend: Fast large

language model serving for RAG with cached knowledge fusion. In *Proceedings of the Twentieth European Conference on Computer Systems*, New York, NY, USA, 2025. Association for Computing Machinery. doi: 10.1145/3689031.3696098. URL <https://www.microsoft.com/en-us/research/uploads/prod/2024/09/eurosys25-final999.pdf>.

Yu, S., Shu, J., Ni, Y., Qian, K., Li, X., Wang, Y., Zhang, J., Xu, Z., Yang, S., Zhu, L., Zhai, E., Lu, Q., Xing, J., Lu, Y., Jin, X., Liu, X., and Xu, H. Pythia: Exploiting workflow predictability for efficient agent-native LLM serving, 2026. URL <https://arxiv.org/abs/2604.25899>.

Zhang, R., Kim, C., Feng, S., Du, K., Liu, Y., Zhong, Y., Ching, C.-W., Jiang, J., and Hu, L. Learning agent execution for kv-cache management in agentic serving, 2026.

Zheng, H., Fu, F., Wu, J., Yuan, B., Zhang, Y., Wang, H., Zhu, Y., Yan, X., and Jiang, J. Efficient serving for dynamic agent workflows with prediction-based KV-cache management, 2026. URL <https://arxiv.org/abs/2605.06472>.

Zheng, L., Yin, L., Xie, Z., Sun, C., Huang, J., Yu, C. H., Cao, S., Kozyrakis, C., Stoica, I., Gonzalez, J. E., Barrett, C., and Sheng, Y. Sglang: Efficient execution of structured language model programs. In *Advances in Neural Information Processing Systems*, volume 37, 2024.